

Manual Book

Website Chatbot Fisika Berbasis JavaScript dan Groq API

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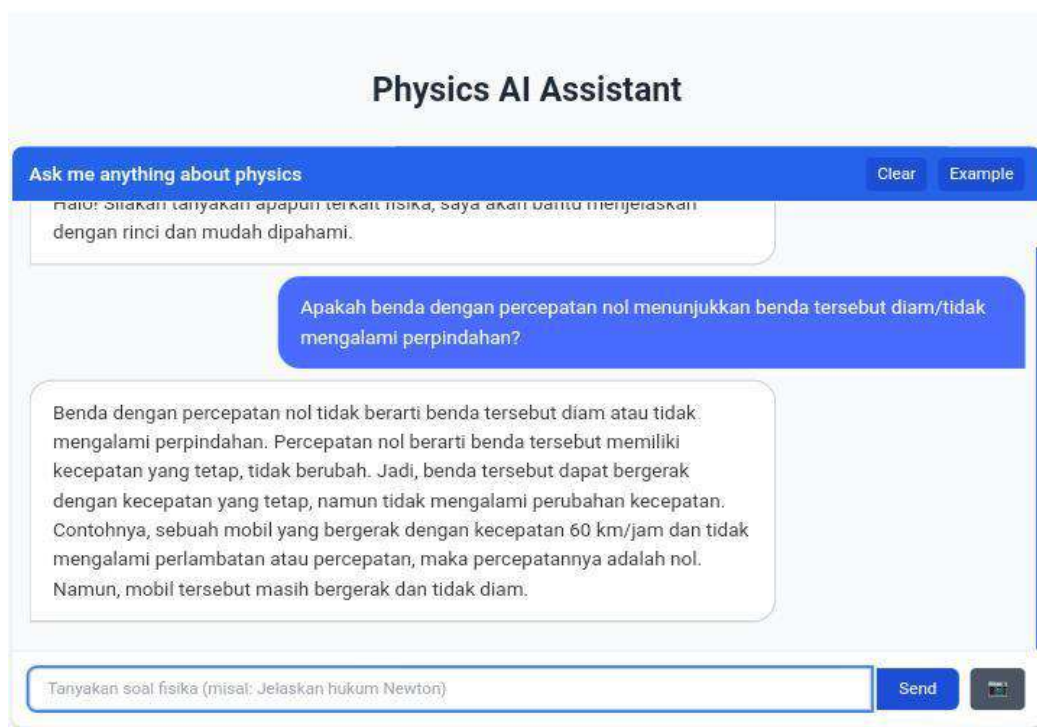
Novina Suryaningtyas

Ahda Zarir Rohman

Langkah Penggunaan

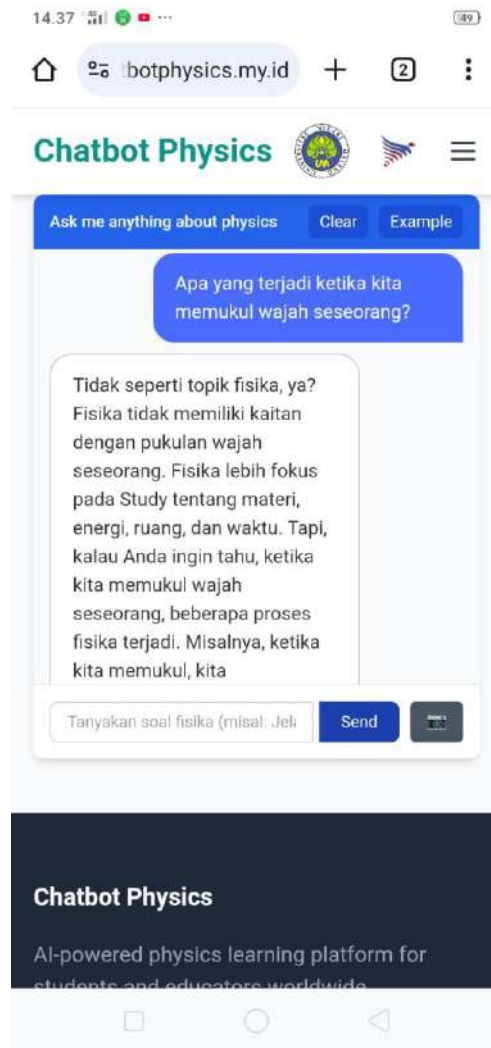
1. Buka browser dan akses URL: chatbotphysics.my.id
2. Pada tampilan utama, pengguna akan melihat jendela chatbot.
3. Ketikkan pertanyaan terkait materi fisika ke dalam kolom pesan.
4. Klik tombol 'Kirim' atau tekan Enter untuk mengirim pesan.
5. Chatbot akan merespons secara otomatis berdasarkan input pengguna.
6. Gunakan fitur percakapan untuk mengeksplorasi berbagai konsep fisika.

Tampilan Aplikasi (Screenshots)



Gambar 1. Tampilan Chatbot Fisika dengan Desktop

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Gambar 2. Tampilan Chatbot Fisika dengan Smartphone

About Chatbot Physics

Chatbot Physics is an advanced AI-powered educational platform designed to revolutionize how students learn physics. Our intelligent assistant combines cutting-edge artificial intelligence with comprehensive physics knowledge to provide instant, accurate, and educational solutions to physics problems of all complexity levels.

Our Mission

Making physics accessible and understandable for students, educators, and enthusiasts worldwide.

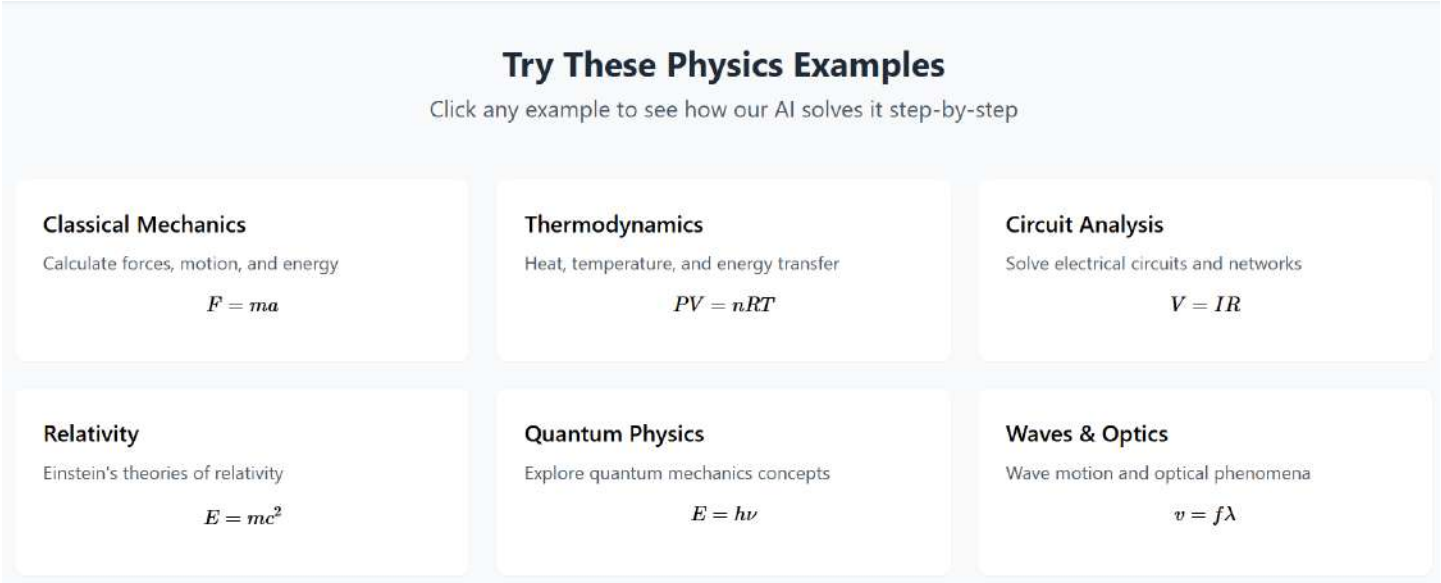
Our Technology

Powered by advanced AI models trained on millions of physics problems and scientific literature.

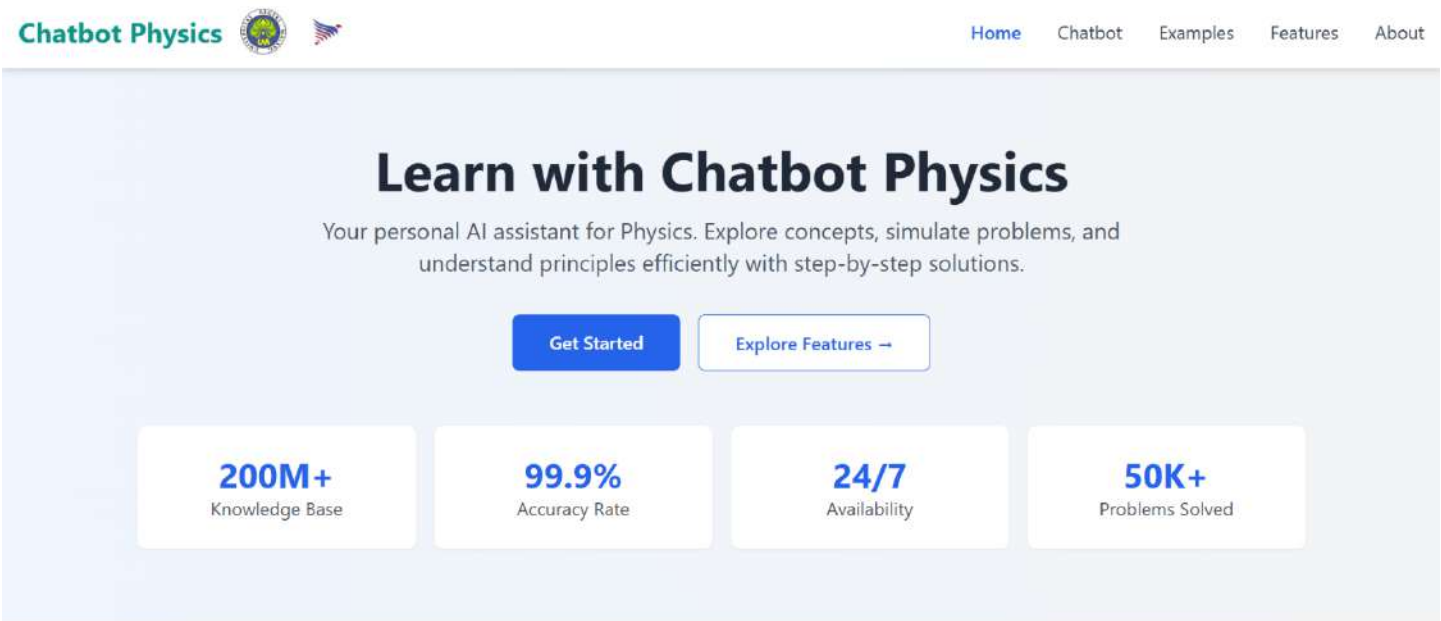
Our Vision

Creating the future of physics education through intelligent, interactive, and personalized learning.

Gambar 3. Laman Tentang Chatbot Fisika



Gambar 4. Laman Contoh Persamaan Fisika



Gambar 5. Laman Utama Chatbot Fisika

Potongan Source Code

ScriptJs

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Chatbot Physics - AI-Powered Physics
Learning</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <script
src="https://polyfill.io/v3/polyfill.min.js?features=es6">
</script>
  <script id="MathJax-script" async
src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-
mml-cthtml.js"></script>
  <style>
    .chatbox {
      scrollbar-width: thin;
      scrollbar-color: #4a6bff #f1f1f1;
    }
    .chatbox::-webkit-scrollbar {
      width: 8px;
    }
    .chatbox::-webkit-scrollbar-thumb {
      background-color: #4a6bff;
      border-radius: 4px;
    }
    .user-message {
      align-self: flex-end;
      background-color: #4a6bff;
      color: white;
      padding: 0.75rem 1rem;
      border-radius: 1rem 1rem 0 1rem;
      max-width: 75%;
      margin-bottom: 0.5rem;
      font-size: 0.875rem; /* Ubah ukuran di sini */
    }

    .bot-message {
      align-self: flex-start;
      background-color: white;
      color: #333;
      border: 1px solid #ccc;
      padding: 0.75rem 1rem;
      border-radius: 1rem 1rem 1rem 0;
      max-width: 75%;
      margin-bottom: 0.5rem;
      font-size: 0.875rem; /* Ubah ukuran di sini */
    }
```

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```
}
.bot-message img, .user-message img {
  max-width: 100%;
  height: auto;
  margin-top: 0.5rem;
  border-radius: 0.5rem;
}

</style>
</head>
<body class="bg-gray-50">
  <!-- Header -->
  <header class="bg-white shadow-md sticky top-0 z-50">
    <nav class="container mx-auto px-4 py-3 flex justify-between items-center">
      <div class="flex items-center space-x-4">
        <h1 class="text-2xl font-bold text-teal-600">Chatbot Physics</h1>
        
        
      </div>
      <ul class="hidden md:flex space-x-8">
        <li><a href="#" class="text-blue-600 font-medium">Home</a></li>
        <li><a href="#chatbot" class="text-gray-700 hover:text-blue-600">Chatbot</a></li>
        <li><a href="#examples" class="text-gray-700 hover:text-blue-600">Examples</a></li>
        <li><a href="#features" class="text-gray-700 hover:text-blue-600">Features</a></li>
        <li><a href="#about" class="text-gray-700 hover:text-blue-600">About</a></li>
      </ul>
      <button class="md:hidden text-gray-700">
        <svg xmlns="http://www.w3.org/2000/svg" class="h-6 w-6" fill="none" viewBox="0 0 24 24" stroke="currentColor">
          <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M4 6h16M4 12h16M4 18h16" />
        </svg>
      </button>
    </nav>
  </header>

  <!-- Hero Section -->
  <section class="bg-gradient-to-r from-blue-50 to-gray-100 py-16">
    <div class="container mx-auto px-4 text-center">
      <h1 class="text-4xl md:text-5xl font-bold text-
```

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Learn with Chatbot Physics

Your personal AI assistant for Physics.

Explore concepts, simulate problems, and understand principles efficiently with step-by-step solutions.

[Get Started](#)

[Explore Features →](#)

200M+

Knowledge Base

99.9%

Accuracy Rate

24/7

Availability

50K+

Problems Solved

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```
<section id="features" class="py-16 bg-white">
  <div class="container mx-auto px-4">
    <h2 class="text-3xl font-bold text-center text-
gray-800 mb-2">Advanced AI Features</h2>
    <p class="text-xl text-gray-600 text-center max-
w-3xl mx-auto mb-12">
      Powered by cutting-edge artificial intelligence
      technology
    </p>
    <div class="grid grid-cols-1 md:grid-cols-2
lg:grid-cols-3 gap-8">
      <div class="bg-gray-50 p-6 rounded-lg
shadow-sm hover:shadow-md transition-shadow">
        <div class="text-4xl mb-4"> 📖 </div>
        <h3 class="text-xl font-semibold mb-
2">Step-by-Step Solutions</h3>
        <p class="text-gray-600">
          Get detailed explanations with
          mathematical steps and conceptual understanding for
          every physics problem.
        </p>
      </div>
      <div class="bg-gray-50 p-6 rounded-lg
shadow-sm hover:shadow-md transition-shadow">
        <div class="text-4xl mb-4"> 📷 </div>
        <h3 class="text-xl font-semibold mb-
2">Image Recognition</h3>
        <p class="text-gray-600">
          Upload photos of physics problems,
          equations, or circuit diagrams for instant AI analysis and
          solutions.
        </p>
      </div>
      <div class="bg-gray-50 p-6 rounded-lg
shadow-sm hover:shadow-md transition-shadow">
        <div class="text-4xl mb-4"> Σ </div>
        <h3 class="text-xl font-semibold mb-
2">Math Rendering</h3>
        <p class="text-gray-600">
          Professional LaTeX equation display
          with proper mathematical formatting for complex
          physics formulas.
        </p>
      </div>
      <div class="bg-gray-50 p-6 rounded-lg
shadow-sm hover:shadow-md transition-shadow">
        <div class="text-4xl mb-4"> ⚡ </div>
        <h3 class="text-xl font-semibold mb-
2">Circuit Analysis</h3>
        <p class="text-gray-600">
          Specialized tools for analyzing electrical
          circuits, resistor networks, and electronic components.
        </p>
      </div>
    </div>
  </div>
```


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</div>

<div class="bg-gray-50 p-6 rounded-lg shadow-sm hover:shadow-md transition-shadow">

<div class="text-4xl mb-4">↔</div>

<h3 class="text-xl font-semibold mb-2">Multiple Methods</h3>

<p class="text-gray-600">

Learn different approaches to solve the same problem with alternative solution methods and techniques.

</p>

</div>

<div class="bg-gray-50 p-6 rounded-lg shadow-sm hover:shadow-md transition-shadow">

<div class="text-4xl mb-4">🎓</div>

<h3 class="text-xl font-semibold mb-2">Educational Guidance</h3>

<p class="text-gray-600">

Focus on learning with conceptual explanations, practice recommendations, and study tips.

</p>

</div>

</div>

</div>

</section>

<!-- Examples Section -->

<section id="examples" class="py-16 bg-gray-50">

<div class="container mx-auto px-4">

<h2 class="text-3xl font-bold text-center text-gray-800 mb-2">Try These Physics Examples</h2>

<p class="text-xl text-gray-600 text-center max-w-3xl mx-auto mb-12">

Click any example to see how our AI solves it step-by-step

</p>

<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">

<div class="bg-white p-6 rounded-lg shadow-sm hover:shadow-md transition-shadow cursor-pointer" onclick="useExample('Explain Newton\\'s Second Law (F=ma)')">

<h3 class="text-xl font-semibold mb-2">Classical Mechanics</h3>

<p class="text-gray-600 mb-4">Calculate forces, motion, and energy</p>

<div class="formula">[$F = ma$]</div>

</div>

<div class="bg-white p-6 rounded-lg shadow-sm hover:shadow-md transition-shadow cursor-pointer" onclick="useExample('Explain the Ideal Gas Law (PV=nRT)')">

<h3 class="text-xl font-semibold mb-2">Thermodynamics</h3>

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```
<p class="text-gray-600 mb-4">Heat,
temperature, and energy transfer</p>
<div class="formula">[ PV = nRT \]</div>
</div>
<div class="bg-white p-6 rounded-lg shadow-
sm hover:shadow-md transition-shadow cursor-pointer"
onclick="useExample('Explain Ohm\\'s Law (V=IR)')">
  <h3 class="text-xl font-semibold mb-
2">Circuit Analysis</h3>
  <p class="text-gray-600 mb-4">Solve
electrical circuits and networks</p>
  <div class="formula">[ V = IR \]</div>
</div>
<div class="bg-white p-6 rounded-lg shadow-
sm hover:shadow-md transition-shadow cursor-pointer"
onclick="useExample('Explain Einstein\\'s mass-energy
equivalence (E=mc^2)')">
  <h3 class="text-xl font-semibold mb-
2">Relativity</h3>
  <p class="text-gray-600 mb-4">Einstein's
theories of relativity</p>
  <div class="formula">[ E = mc^2 \]</div>
</div>
<div class="bg-white p-6 rounded-lg shadow-
sm hover:shadow-md transition-shadow cursor-pointer"
onclick="useExample('Explain Planck\\'s relation for
photon energy (E=hv)')">
  <h3 class="text-xl font-semibold mb-
2">Quantum Physics</h3>
  <p class="text-gray-600 mb-4">Explore
quantum mechanics concepts</p>
  <div class="formula">[ E = h\nu \]</div>
</div>
<div class="bg-white p-6 rounded-lg shadow-
sm hover:shadow-md transition-shadow cursor-pointer"
onclick="useExample('Explain the wave equation
(v=f\lambda)')">
  <h3 class="text-xl font-semibold mb-
2">Waves & Optics</h3>
  <p class="text-gray-600 mb-4">Wave
motion and optical phenomena</p>
  <div class="formula">[ v = f\lambda
\]</div>
</div>
</div>
</div>
</section>
```

```
<!-- About Section -->
<section id="about" class="py-16 bg-white">
  <div class="container mx-auto px-4">
    <h2 class="text-3xl font-bold text-center text-
gray-800 mb-2">About Chatbot Physics</h2>
    <p class="text-xl text-gray-600 text-center max-
```

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w-4xl mx-auto mb-12">

Chatbot Physics is an advanced AI-powered educational platform designed to revolutionize how students learn physics. Our intelligent assistant combines cutting-edge artificial intelligence with comprehensive physics knowledge to provide instant, accurate, and educational solutions to physics problems of all complexity levels.

</p>

<div class="grid grid-cols-1 md:grid-cols-3 gap-8">

<div class="bg-gray-50 p-6 rounded-lg shadow-sm text-center">

<h3 class="text-xl font-semibold mb-4">Our Mission</h3>

<p class="text-gray-600">

Making physics accessible and understandable for students, educators, and enthusiasts worldwide.

</p>

</div>

<div class="bg-gray-50 p-6 rounded-lg shadow-sm text-center">

<h3 class="text-xl font-semibold mb-4">Our Technology</h3>

<p class="text-gray-600">

Powered by advanced AI models trained on millions of physics problems and scientific literature.

</p>

</div>

<div class="bg-gray-50 p-6 rounded-lg shadow-sm text-center">

<h3 class="text-xl font-semibold mb-4">Our Vision</h3>

<p class="text-gray-600">

Creating the future of physics education through intelligent, interactive, and personalized learning.

</p>

</div>

</div>

</div>

</section>

<!-- Chatbot Section -->

<section id="chatbot" class="py-10 bg-gray-50">

<div class="container mx-auto px-4">

<h2 class="text-2xl font-bold text-center text-gray-800 mb-6">Physics AI Assistant</h2>

<div class="max-w-4xl mx-auto bg-white rounded-lg shadow-md overflow-hidden">

<div class="bg-blue-600 text-white px-3 py-2 flex justify-between items-center">

<h3 class="text-xs sm:text-sm font-medium">Ask

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me anything about physics</h3>

```
<div class="flex space-x-2">
  <button id="clear-chat" class="bg-blue-700
hover:bg-blue-800 text-white px-2 py-1 rounded text-
xs">
    Clear
  </button>
  <button id="example-btn" class="bg-blue-700
hover:bg-blue-800 text-white px-2 py-1 rounded text-
xs">
    Example
  </button>
</div>
</div>
```

```
<div id="chatbox" class="chatbox h-80 p-3
overflow-y-auto bg-gray-50 flex flex-col">
  <!-- Messages will appear here -->
</div>
```

```
<div class="p-3 bg-white border-t border-gray-200
flex gap-1">
  <input type="text" id="userInput"
placeholder="Tanyakan soal fisika (misal: Jelaskan
hukum Newton)"
class="flex-1 border border-gray-300 rounded-
l-md px-3 py-1.5 text-xs focus:outline-none focus:ring-2
focus:ring-blue-500">
  <button id="send-btn" class="bg-blue-700
hover:bg-blue-800 text-white px-4 py-1.5 rounded-r-md
text-xs">
    Send
  </button>
  <label for="image-upload" class="ml-1 bg-gray-
600 hover:bg-gray-700 text-white px-3 py-1.5 rounded
cursor-pointer text-xs">
    
  </label>
  <input type="file" id="image-upload"
accept="image/*" class="hidden">
</div>
</div>
</div>
</section>
```

```
<!-- Footer -->
<footer class="bg-gray-800 text-white pt-12 pb-6">
  <div class="container mx-auto px-4">
    <div class="grid grid-cols-1 md:grid-cols-4 gap-
8 mb-8">
      <div>
        <h3 class="text-lg font-semibold mb-
4">Chatbot Physics</h3>
        <p class="text-gray-400">
```

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AI-powered physics learning platform for students and educators worldwide.

```
</p>
</div>
<div>
  <h3 class="text-lg font-semibold mb-4">Features</h3>
  <ul class="space-y-2">
    <li><a href="#features" class="text-gray-400 hover:text-white">AI Problem Solver</a></li>
    <li><a href="#examples" class="text-gray-400 hover:text-white">Example Problems</a></li>
    <li><a href="#features" class="text-gray-400 hover:text-white">Advanced Features</a></li>
  </ul>
</div>
<div>
  <h3 class="text-lg font-semibold mb-4">Resources</h3>
  <ul class="space-y-2">
    <li><a href="#" class="text-gray-400 hover:text-white">User Guide</a></li>
    <li><a href="#examples" class="text-gray-400 hover:text-white">Physics Topics</a></li>
    <li><a href="#" class="text-gray-400 hover:text-white">Study Materials</a></li>
  </ul>
</div>
<div>
  <h3 class="text-lg font-semibold mb-4">Support</h3>
  <ul class="space-y-2">
    <li><a href="#" class="text-gray-400 hover:text-white">Help Center</a></li>
    <li><a href="#" class="text-gray-400 hover:text-white">Contact Us</a></li>
    <li><a href="#" class="text-gray-400 hover:text-white">Privacy Policy</a></li>
  </ul>
</div>
</div>
<div class="border-t border-gray-700 pt-6 flex flex-col md:flex-row justify-between items-center">
  <p class="text-gray-400 mb-4 md:mb-0">© 2025 Chatbot Physics. All rights reserved.</p>
  <p id="current-time" class="text-gray-400">21:46 14/06/2025</p>
</div>
</div>
</footer>
```

</html>

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